

Evaluating Multimodal Alignment in ChatGPT and Gemini: An Empirical Study of Inter- and Intra-Model Consistency in Image Description and Generation

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Abstract—This study examines how well three perceptual metrics—LPIPS, CLIP cosine similarity, and MS-SSIM—align with human preferences over image variants generated by ChatGPT (DALL-E) and Gemini for 15 reference images. Ten raters ranked four variants per reference image. Inter-rater agreement (Kendall’s W) is computed, automatic scores are summarized, and the correlation between human mean ranks and each automatic metric is measured. Humans show moderate agreement ($\bar{W} = 0.40$). Across variants, automatic metrics favor different candidates, and only CLIP shows a modest positive correlation with human ranks (mean Spearman 0.36). LPIPS and MS-SSIM correlate weakly. These findings suggest that single off-the-shelf metrics are insufficient proxies for human judgments on this task and that model selection should incorporate targeted human feedback.

Index Terms—human evaluation, perceptual metrics, LPIPS, CLIP, MS-SSIM, alignment

I. INTRODUCTION

Text-to-image systems are typically benchmarked with automatic similarity scores, yet deployment decisions rely on human perception. This study investigates how three widely used metrics align with human preferences over 15 sets of four image variants each. The contributions are: (i) a paired human study with ten raters, (ii) a unified pipeline to compute LPIPS, CLIP, and MS-SSIM on the same data, and (iii) an empirical comparison of metric rankings against human rankings and inter-rater agreement. Key findings are summarized and visual evidence is provided to illustrate where metrics and humans diverge.

Two research questions structure the study:

- RQ1 How consistent, with respect to human preferences, are multimodal AI models in aligning descriptions and generations, both within the same model and across different models?
- RQ2 How well do automatic multimodal evaluation metrics align with human judgments?

Related work on human alignment for generative models often focuses on text-only LLMs or aesthetic predictors; fewer studies contrast multiple perceptual metrics against ground-truth human orderings for images. By keeping the study small and controlled (15 prompts, fixed four-way choice),

the behavior of off-the-shelf metrics is isolated without confounds from model sampling stochasticity. Visual inspection is emphasized alongside quantitative summaries to highlight qualitative failure modes.

II. RELATED WORK

Perceptual similarity has a long history in image quality assessment, from SSIM and MS-SSIM [3], [4] to learned metrics such as LPIPS [1]. CLIP-based similarity has become a common proxy for text-image alignment and aesthetics [2], [5]. Recent work on human preference modeling for image generation often trains reward or aesthetic models, yet many practitioners still rely on these off-the-shelf metrics. This study complements prior work by directly contrasting human rank lists with three popular metrics on the same set of variants. Unlike large-scale benchmarks that aggregate thumbnails, full-resolution pairs and a strict four-way ranking are used to reduce ties. Broader surveys on multimodal LLMs outline architectures and alignment strategies [7], [8]. CLIP-based image quality assessment variants such as CLIP-AGIQA extend automated evaluation for generated images [9], while LLM-based evaluators offer another automatic preference signal [10]. Representation alignment work (e.g., CLIP embeddings [2]) and cycle-consistency ideas motivate comparing description and generation consistency. In contrast to metric- or model-focused studies, the present work emphasizes comparative analysis of two AI models and their intra-model consistency using both human rankings and automatic metrics on the same bespoke dataset. Other quantitative measures such as FID [6] or aesthetic predictors provide dataset-level signals, but they do not directly reflect the relative ranking of small candidate sets shown to end users. The analysis therefore focuses on the per-prompt, per-variant view that matches typical human-in-the-loop selection.

III. DATA AND EXPERIMENTAL SETUP

A. Image Variants

Each of the 15 images has a reference version and four model-generated variants (a)–(d). Variants differ in composition and style but share the same prompt intent. Across

prompts, variation is observed in color palette, object placement, and adherence to textual details, providing a useful stress test for perceptual metrics. Reference images were collected specifically for this study to minimize the chance that either model encountered them during pretraining. The small bespoke set spans natural elements (landscapes, weather), embedded text, everyday objects, and people, ensuring diversity in content types that stress both compositional and semantic alignment. Letters (a)–(d) encode which AI model generated the image and which model wrote the caption it used. Variant (a) is generated by Gemini based on a caption written by Gemini itself. Variant (b) is generated by Gemini using a caption written by ChatGPT. Variant (c) is generated by ChatGPT/DALL–E using a caption written by Gemini. Variant (d) is generated by ChatGPT/DALL–E using a caption written by ChatGPT. Variants (a)/(b) and (c)/(d) therefore differ only in caption source; (a)/(c) and (b)/(d) contrast the two generators under the same caption.

B. Human Evaluation

Ten raters (balanced male/female) ranked the four variants for every image. The split enables reporting per-group heatmaps. Inter-rater agreement is summarized with Kendall’s W per image and averaged. Raters saw all four variants side-by-side and selected a strict ordering after brief instructions about prioritizing semantic fidelity and overall appeal.

C. Automatic Metrics

LPIPS (AlexNet backbone), MS-SSIM, and CLIP cosine similarity (ViT-B/32) are computed. Images are resized to match references for perceptual metrics. Per-variant means and per-image “winners” (best score per metric) are reported.

D. Outputs and Reproducibility

All intermediate CSVs and plots (metric means, human vs. automatic, rank heatmaps, correlation candles) are produced by the analysis pipeline provided with this study. No external assets beyond the repository images are required.

E. Evaluation Protocol

For each image, raters view the four candidates simultaneously and rank them 1–4. Top-1 frequencies are aggregated and Kendall’s W is computed to quantify agreement. For automatic metrics, image sizes are aligned to the reference, scores are computed, and variants are ranked per metric and image. Correlations between human mean ranks and metric ranks are reported per image and averaged across all 15 images.

F. Implementation Details

Images are resized to the reference dimensions before computing LPIPS and MS-SSIM. CLIP uses a ViT-B/32 backbone and standard preprocessing. Random seeds are fixed where applicable; no sampling randomness is introduced because all images are pre-generated.

IV. RESULTS

A. Human Preferences

Top-1 choice rates are: (a) (44%), (b) (22%), (c) (13.3%), (d) (20.7%). Inter-rater agreement across all images averages $W = 0.40$ (Fig. 2), indicating moderate but not strong consensus. Image-level W ranges from low agreement to near-consensus, suggesting prompt-dependent ambiguity. Male and female subgroup heatmaps are omitted because they closely mirror the overall distribution; both subgroups emphasize variant (a) with only minor positional shifts. Fig. 1 summarizes placement percentages across all raters. Given the mapping, raters most often preferred Gemini-generated images prompted by Gemini captions (a), followed by Gemini with ChatGPT captions (b), and least often ChatGPT-generated images conditioned on Gemini captions (c). The ChatGPT stack with ChatGPT captions (d) lands between (b) and (c), suggesting both the generator and caption source influence perceived quality.

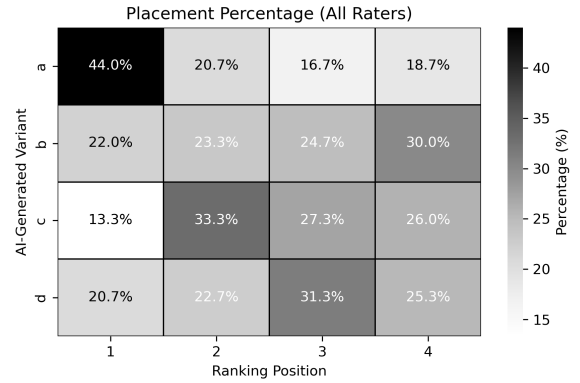


Fig. 1. Human placement percentages across all raters (grayscale heatmap). Variant (a) is most often ranked first.

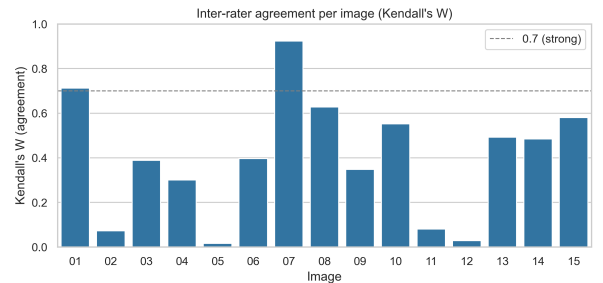


Fig. 2. Per-image inter-rater agreement (Kendall’s W) with mean $W = 0.40$ indicated.

B. Metric Outcomes

No single metric matches the human-favored variant. Visual inspection confirms that LPIPS and MS-SSIM can reward higher-frequency detail even when humans prefer semantic fidelity, while CLIP occasionally overweights textual alignment. Mean metric scores per variant (lower LPIPS is better; higher

MS-SSIM/CLIP is better) are reported in Fig. 3, while Fig. 4 shows how often each variant wins per metric. When counting per-image wins (global percentage over all rows), LPIPS favors (a) (11.7%) while CLIP favors (a) (10.0%) and (d) (8.3%); MS-SSIM splits wins across (b)–(d) (8.3% each) with minimal wins for (a) (1.7%). LPIPS and CLIP lean toward the Gemini-generated images prompted by Gemini captions variant (a), whereas MS-SSIM spreads wins across Gemini-generated images prompted by ChatGPT captions (b) and ChatGPT-generated images prompted by ChatGPT captions (d), indicating that neither Gemini nor ChatGPT pipelines dominate across perceptual criteria. The spread reinforces that off-the-shelf metrics disagree on whether Gemini-generated or ChatGPT-generated images better match references, even when fed the same caption source. Fig. 5 contrasts human top-1 rates with global metric win percentages to highlight where automatic selection diverges from human preference. Across metrics, the mean scores in Fig. 3 lie in a narrow band, implying that automatic measures view the variants as roughly similar, whereas human preferences are more polarized toward variant (a), underscoring a gap between numerical similarity and subjective choice.

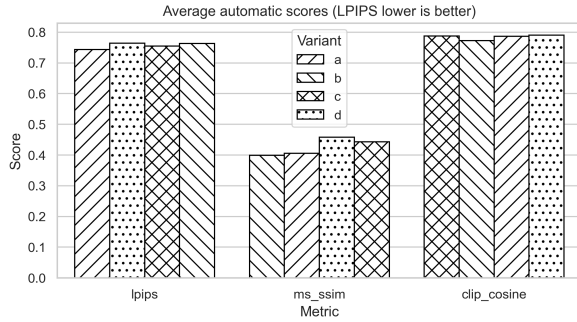


Fig. 3. Mean automatic scores per variant. LPIPS lower is better; MS-SSIM/CLIP higher is better.

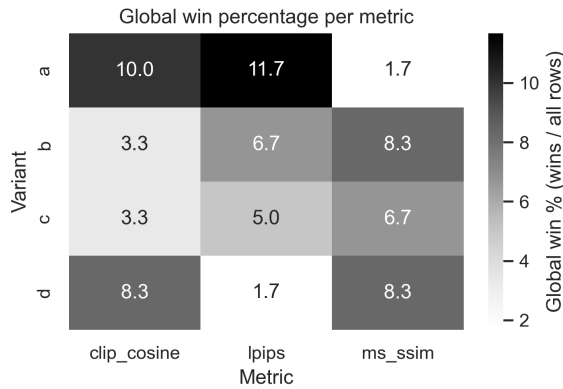


Fig. 4. Global percentage of per-image wins by variant and metric.

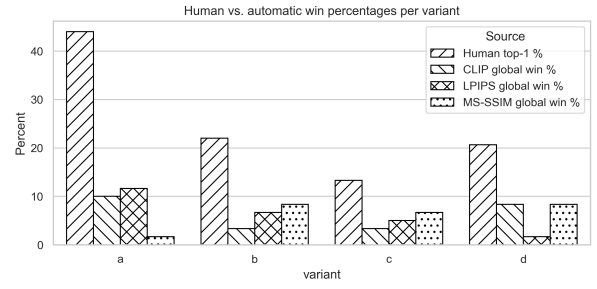


Fig. 5. Human top-1 rates compared to automatic global win percentages per variant.

C. Rank Distributions

Beyond single winners, the typical rank positions of each variant are examined when ordered by a metric. LPIPS places variant (a) first most often but also pushes it to lower ranks in some prompts. MS-SSIM distributes ranks more evenly, while CLIP concentrates wins on (a) and (d). These distributions echo the weak correlations with human preferences: no metric consistently mirrors the human ordering.

D. Correlation Analysis

Table I reports the mean correlation between human mean ranks and each metric over 15 images. Only CLIP shows a modest positive relationship; LPIPS and MS-SSIM are weak. The candle plot in Fig. 6 visualizes mean and range (min/max) across Kendall’s Tau, Spearman, and Pearson per image, showing substantial variability across prompts. Some prompts exhibit negative or near-zero correlation, indicating cases where metrics are anti-aligned with human preference.

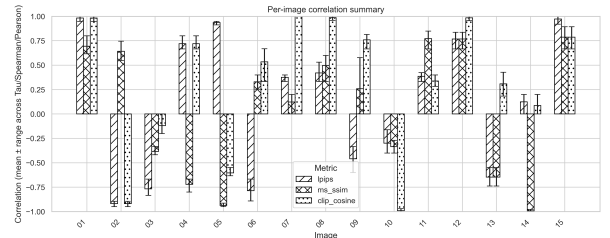


Fig. 6. Per-image correlation summary: mean and range across Kendall, Spearman, and Pearson for each metric (bars with whiskers).

TABLE I
MEAN CORRELATIONS (15 IMAGES) BETWEEN HUMAN MEAN RANKS AND AUTOMATIC METRICS.

Metric	Kendall	Spearman	Pearson
CLIP cosine	0.32	0.36	0.35
LPIPS	0.12	0.11	0.13
MS-SSIM	0.01	0.06	0.10

V. DISCUSSION

Humans prefer variant (a) nearly twice as often as (b) and three times as often as (c), yet automatic metrics do not unanimously select (a). CLIP tracks human rankings

better than LPIPS or MS-SSIM, but still leaves substantial disagreement. Low inter-rater agreement ($W = 0.40$) further limits how strongly any metric can align, suggesting that prompt difficulty and subjective taste both contribute noise. The inconsistency across metrics underscores the need for either metric ensembling, human-in-the-loop ranking, or task-specific fine-tuning. Group-level differences were small in the (omitted) subgroup heatmaps, but the study is underpowered to claim demographic effects; expanding rater diversity and scale is future work. Practical guidance: if constrained to a single metric, CLIP is the least misaligned here; otherwise, collect a small human validation set and calibrate a weighted combination.

A. Qualitative Observations

Visual review of images where metrics disagreed with humans reveals two common patterns: (i) LPIPS and MS-SSIM reward fine texture or sharp edges even when the global composition deviates from the reference intent, and (ii) CLIP rewards semantic matches to the prompt even if artifacts or awkward layouts are present. In several prompts, humans prioritized coherence of objects and overall aesthetics rather than strict similarity to the reference, which explains why variants with slightly softer details could win despite lower metric scores.

B. Limitations

The study covers only 15 prompts and four variants per prompt, limiting statistical power and diversity. Rater demographics are narrow (ten participants), so subgroup analysis is indicative only. All metrics are used off-the-shelf; better alignment may be achievable with fine-tuned or task-specific preference models. Images are evaluated at a single resolution and device configuration; different scales or hardware could modestly shift scores.

C. Practical Implications

For small-batch creative workflows where a designer must pick one of several candidates, these results suggest combining at least one semantic metric (e.g., CLIP) with a quick human pass rather than relying on purely perceptual scores. When human time is scarce, a lightweight calibration set of human rankings can be used to learn metric weights that better approximate local taste. In automated pipelines, exposing multiple candidates to end users may be preferable to auto-selecting a “best” image based solely on a single metric.

D. Future Directions

Expanding the prompt set and rater pool would allow finer-grained subgroup analysis and statistical testing. Additional metrics such as aesthetic predictors, caption-based rewards, or task-conditioned vision-language models could be evaluated. Training a small learned ranker on the collected human data could indicate how much data is needed to surpass off-the-shelf metrics. Finally, testing robustness across resolutions and devices would clarify how sensitive the metrics are to deployment conditions. Prompt engineering is another avenue: future

experiments can explicitly provide models with information about the target audience and task purpose when generating captions or images. For captioning, the audience cue can specify whether the consumer is the same model that will later generate images or a different AI model. For generation, prompts can state whether the output is destined for human ranking or an automatic tool, enabling evaluation of how audience- and purpose-aware instructions affect alignment.

VI. CONCLUSION

This study collected 15 bespoke reference images with four variants each, generated by Gemini and ChatGPT using both self-authored and cross-authored captions, to minimize pretraining leakage and diversify content (people, objects, text, nature). Ten human raters provided a ranking of the four variants for each reference image, yielding moderate inter-rater agreement ($\bar{W} = 0.40$). Automatic evaluation used LPIPS, MS-SSIM, and CLIP to score and rank the same variants, enabling a direct comparison between human preferences and widely used perceptual metrics.

RQ1 Human raters favored Gemini-generated images prompted by Gemini captions (variant (a)), with lower preference for ChatGPT-generated images; intra-model consistency is stronger than inter-model consistency, since each model paired with its own captions ((a) and (d)) is ranked first more often than when using the other model’s captions ((b) and (c)).

RQ2 CLIP shows the strongest (yet modest) alignment with human rankings, while LPIPS and MS-SSIM correlate weakly, so no single automatic metric adequately proxies human judgment.

Across metrics, mean scores cluster tightly, suggesting automatic measures perceive variants as similar, whereas human choices are more polarized toward (a). Selecting variants solely by one metric therefore risks diverging from user preferences. Future work should expand the human study, explore aesthetic and learned preference models, test metric combinations tuned to human-labeled data, and further probe audience- and purpose-aware prompting to improve downstream alignment.

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